

SPRINT FOUR DELIVERABLES

INSTRUCTIONS:

Simply double clicking or running the executable file (Nine Men's Morris.jar) will start the game.

How to Build:

1) After you have opened a terminal/Command Prompt and have navigated to that directory, compile the Engine.java class with

```
javac Engine.java
```

3) This will produce the Engine.class file needed for the JAR file.

4) Next, create a manifest file (saved using the extension .txt) using the text editor and input the following:

Main-Class: Engine

5) Next create the JAR file using this code:

```
jar cfm Nine Men's Morris.jar Manifest.txt Engine.class images Logic  
Manager View
```

6) Run the file:

```
java -jar Engine.jar
```

USER STORIES:

As a/an new player at the student talent exhibition,
I want a tutorial mode in the Nine Men's Morris game,
So that I can learn how to play the game effectively.

As a/an player in the Nine Men's Morris game,
I want the option to toggle "hints" during a match,
So that I can see all the legal moves available for my next move.

Design Rationales

We have incorporated the advanced feature in the Model View Controller pattern established in Sprint 3. An instance of TutorialView is created in the View class, TutorialView contains methods which, when called, present the user with a sequence of images that explain the rules and steps to play the game. Tutorial button provides a clickable button for the user to start the tutorial.

The hint mechanic was implemented by simply utilising the CandidateMgr class, which evaluates valid moves for a player's selected token, based on the context of the board. HintView, provides a draw method which iterates through candidate selections/moves for the current board state and highlights those positions green/yellow accordingly.

Since HintView and TutorialView are instantiated in the View class, they effectively interact with Game the same way BoardView does, and thus belong to the same design pattern structure.

Adding the hint and tutorial features into the application was simple, since we had already created a structure where objects created in View, can influence and interact with Game by individually updating, alongside Game, inside of View. These instances of classes created within View, also have access to the CandidateMgr class that provides methods to return candidate moves. Therefore the new hint and tutorial classes could just be neatly initialised and updated inside of View alongside the other classes.

As we have completed the advanced functions in sprint3, so we didn't do the changes. We fixed problems like no executable instructions found and recommend organising classes in packages.